

# Chapter 5.3 - Cache Basics & Direct-Mapped Caches

Comprehensive Revision Flashcards • 111 Key Concepts & Practice Questions

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| #1 | <b>Q1: What is the general term for any storage managed to take advantage of locality of access?</b>                                  |
|    | <b>Answer:</b> A cache.                                                                                                               |
| #2 | <b>Q2: What is the event called when a processor requests a word that is not in the cache, and what is the immediate result?</b>      |
|    | <b>Answer:</b> It is called a cache miss, and it forces the cache to fetch the requested word from memory.                            |
| #3 | <b>Q3: What is a `direct-mapped cache`?</b>                                                                                           |
|    | <b>Answer:</b> A cache structure in which each memory location is mapped to exactly one location in the cache.                        |
| #4 | <b>Q4: What formula is typically used to map a memory block address to a location in a direct-mapped cache?</b>                       |
|    | <b>Answer:</b> (Block address) modulo (Number of blocks in the cache).                                                                |
| #5 | <b>Q5: How is the modulo operation for address mapping simplified if the number of entries in the cache is a power of 2?</b>          |
|    | <b>Answer:</b> By using the low-order $\log_2(\text{cache size in blocks})$ bits of the address as the index.                         |
| #6 | <b>Q6: What is the purpose of the `tag` field in a cache entry?</b>                                                                   |
|    | <b>Answer:</b> It contains the address information required to identify whether a block in the cache corresponds to a requested word. |
| #7 | <b>Q7: Which portion of the memory address is stored in the cache entry's tag field?</b>                                              |
|    | <b>Answer:</b> The upper portion of the address, corresponding to the bits that are not used as the cache index.                      |
| #8 | <b>Q8: What is the purpose of the `valid bit` in a cache entry?</b>                                                                   |
|    | <b>Answer:</b> It indicates whether the associated block in the cache contains valid data.                                            |
| #9 | <b>Q9: When a processor first starts up, what is the typical state of the cache's valid bits?</b>                                     |
|    | <b>Answer:</b> They are not set (e.g., 'N' or 0), indicating that the cache entries do not contain valid information.                 |

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| #10 | <b>Q10: Caching is perhaps the most important example of what major computer architecture concept?</b>                                                               |
|     | <b>Answer:</b> Prediction.                                                                                                                                           |
| #11 | <b>Q11: Caching relies on the principle of _____ to attempt to find desired data in higher levels of the memory hierarchy.</b>                                       |
|     | <b>Answer:</b> locality                                                                                                                                              |
| #12 | <b>Q12: A referenced memory address is divided into which two primary fields for a direct-mapped cache lookup?</b>                                                   |
|     | <b>Answer:</b> A tag field and a cache index.                                                                                                                        |
| #13 | <b>Q13: What two components of a cache entry uniquely specify the memory address of the word contained in that cache block?</b>                                      |
|     | <b>Answer:</b> The cache index and the tag contents of that block.                                                                                                   |
| #14 | <b>Q14: Why must the total number of entries in a direct-mapped cache be a power of 2?</b>                                                                           |
|     | <b>Answer:</b> Because the index field is used as an address to reference the cache, and an n-bit field has $2^n$ possible values.                                   |
| #15 | <b>Q15: For a cache with a one-word block size, how is a memory address with 64 bits, <math>2^n</math> blocks, and <math>2^m</math> words per block broken down?</b> |
|     | <b>Answer:</b> The tag field is $64 - n - m - 2$ bits, the index is $n$ bits, the block offset is $m$ bits, and the byte offset is 2 bits.                           |
| #16 | <b>Q16: By convention, when a cache is referred to by its size (e.g., a "4 KiB cache"), what components are excluded from this measurement?</b>                      |
|     | <b>Answer:</b> The storage for the tag and valid fields; only the data storage is counted.                                                                           |
| #17 | <b>Q17: For a cache with 64 blocks and a block size of 16 bytes, to what block number does byte address 1200 map?</b>                                                |
|     | <b>Answer:</b> Block number 11, which is calculated as $\lfloor 1200 / 16 \rfloor \bmod 64 = 75 \bmod 64 = 11$ .                                                     |
| #18 | <b>Q18: How do larger block sizes in a cache exploit spatial locality?</b>                                                                                           |
|     | <b>Answer:</b> By fetching adjacent words along with the requested word, they decrease the miss rate for subsequent sequential accesses.                             |

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| #19 | <b>Q19: What is a potential negative consequence of making a cache's block size too large relative to the total cache size?</b>           |
|     | <b>Answer:</b> The miss rate may increase due to a high degree of competition for the few available blocks.                               |
| #20 | <b>Q20: What is the primary drawback of increasing cache block size in terms of performance?</b>                                          |
|     | <b>Answer:</b> It increases the miss penalty because more time is required to transfer the larger block from memory.                      |
| #21 | <b>Q21: What are the two components that determine the time required to fetch a block during a cache miss?</b>                            |
|     | <b>Answer:</b> The latency to the first word and the transfer time for the rest of the block.                                             |
| #22 | <b>Q22: Which general guideline is valid for a cache designer concerning memory bandwidth?</b>                                            |
|     | <b>Answer:</b> The higher the memory bandwidth, the larger the cache block size can be.                                                   |
| #23 | <b>Q23: What is a `cache miss`?</b>                                                                                                       |
|     | <b>Answer:</b> A request for data from the cache that cannot be filled because the data is not present in the cache.                      |
| #24 | <b>Q24: The processing of a cache miss creates a _____ in the processor, freezing the contents of registers while waiting for memory.</b> |
|     | <b>Answer:</b> pipeline stall                                                                                                             |
| #25 | <b>Q25: What is the first step in handling an instruction cache miss?</b>                                                                 |
|     | <b>Answer:</b> Send the original Program Counter (PC) value to the memory system.                                                         |
| #26 | <b>Q26: After sending the address, what is the second step in handling an instruction cache miss?</b>                                     |
|     | <b>Answer:</b> Instruct main memory to perform a read and wait for the access to complete.                                                |
| #27 | <b>Q27: What is the third step in handling an instruction cache miss, after the data is returned from memory?</b>                         |
|     | <b>Answer:</b> Write the data into the cache, write the upper address bits into the tag field, and set the valid bit to on.               |
| #28 | <b>Q28: What is the final step in handling an instruction cache miss?</b>                                                                 |
|     | <b>Answer:</b> Restart the instruction execution, which will refetch the instruction, causing a cache hit.                                |

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| #29 | <b>Q29: If a write operation modifies only the data cache without changing main memory, the cache and memory are said to be ____.</b>            |
|     | <b>Answer:</b> inconsistent                                                                                                                      |
| #30 | <b>Q30: What is the `write-through` scheme for handling writes?</b>                                                                              |
|     | <b>Answer:</b> A scheme in which writes always update both the cache and the next lower level of the memory hierarchy.                           |
| #31 | <b>Q31: What is the primary performance drawback of a simple write-through scheme?</b>                                                           |
|     | <b>Answer:</b> Every write incurs the long latency of an access to main memory, which can significantly slow down the processor.                 |
| #32 | <b>Q32: What is a `write buffer`?</b>                                                                                                            |
|     | <b>Answer:</b> A queue that holds data while it is waiting to be written to memory, allowing the processor to continue execution.                |
| #33 | <b>Q33: When using a write buffer, under what condition must the processor stall during a write operation?</b>                                   |
|     | <b>Answer:</b> The processor must stall if the write buffer is full.                                                                             |
| #34 | <b>Q34: What is the `write-back` scheme for handling writes?</b>                                                                                 |
|     | <b>Answer:</b> A scheme where a write updates only the block in the cache, and the modified block is written to memory only when it is replaced. |
| #35 | <b>Q35: What is the `write allocate` policy for a write miss?</b>                                                                                |
|     | <b>Answer:</b> The block is fetched from memory and placed into the cache, and then the relevant word is overwritten.                            |
| #36 | <b>Q36: What is the `no write allocate` policy for a write miss?</b>                                                                             |
|     | <b>Answer:</b> The word is updated in main memory, but the block is not brought into the cache.                                                  |
| #37 | <b>Q37: What is the primary motivation for using a `no write allocate` policy?</b>                                                               |
|     | <b>Answer:</b> To avoid unnecessary memory fetches when a program writes an entire block of data, such as when an OS zeros a page.               |

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| #38 | <b>Q38: Why is implementing stores more complex in a write-back cache compared to a write-through cache?</b>                                                 |
|     | <b>Answer:</b> The cache block cannot be overwritten until a hit is confirmed, because memory does not have a correct copy of the potentially modified data. |
| #39 | <b>Q39: What is a `split cache`?</b>                                                                                                                         |
|     | <b>Answer:</b> A memory hierarchy level composed of two independent caches, one for instructions and one for data, that operate in parallel.                 |
| #40 | <b>Q40: What is the main advantage of using a split instruction and data cache?</b>                                                                          |
|     | <b>Answer:</b> It doubles the cache bandwidth by allowing simultaneous instruction and data accesses.                                                        |
| #41 | <b>Q41: What is a potential disadvantage of a split cache compared to a single combined cache of the same total size?</b>                                    |
|     | <b>Answer:</b> The effective miss rate can be slightly higher due to the rigid partitioning of cache entries.                                                |
| #42 | <b>Q42: What is the `early restart` technique used for?</b>                                                                                                  |
|     | <b>Answer:</b> To reduce miss penalty by resuming execution as soon as the requested word of a block arrives, without waiting for the entire block.          |
| #43 | <b>Q43: What is the `critical word first` technique?</b>                                                                                                     |
|     | <b>Answer:</b> A scheme to reduce miss penalty where the memory system transfers the requested word first, then sends the rest of the block.                 |
| #44 | <b>Q44: When a cache miss requires replacing an existing block in a direct-mapped cache, what principle of locality is being exploited?</b>                  |
|     | <b>Answer:</b> Temporal locality, as a recently referenced item replaces a less recently referenced item.                                                    |
| #45 | <b>Q45: In the provided example, why does a reference to address 18 (10010two) replace the entry for address 26 (11010two)?</b>                              |
|     | <b>Answer:</b> Because both memory addresses map to the same cache index (010two) in the direct-mapped cache.                                                |
| #46 | <b>Q46: In the example of the 4 KiB cache in Figure 5.10, how many bits of the 64-bit address are used for the tag?</b>                                      |
|     | <b>Answer:</b> 52 bits, calculated as 64 total bits - 10 index bits - 2 byte-offset bits.                                                                    |

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| #47 | <b>Q47: What two conditions must be met for a cache access to result in a hit?</b>                                                                                     |
|     | <b>Answer:</b> The tag in the cache entry must match the tag portion of the address, and the entry's valid bit must be on.                                             |
| #48 | <b>Q48: How can a write buffer also be used with a write-back cache to reduce miss penalty?</b>                                                                        |
|     | <b>Answer:</b> When a miss replaces a modified block, the modified block can be moved to the buffer to be written to memory later, while the requested block is read.  |
| #49 | <b>Q49: How is the full memory address 10010two reconstructed from the cache entry at index 010two in Figure 5.9f?</b>                                                 |
|     | <b>Answer:</b> By concatenating the tag field (10two) with the index field (010two).                                                                                   |
| #50 | <b>Q50: In the context of computer memory hierarchy, what is a cache?</b>                                                                                              |
|     | <b>Answer:</b> A small, fast storage level that acts as a safe place to store things (data) needed by the processor, positioned between the processor and main memory. |
| #51 | <b>Q51: The term 'cache' is used to refer to any storage managed to take advantage of ____.</b>                                                                        |
|     | <b>Answer:</b> locality of access                                                                                                                                      |
| #52 | <b>Q52: What is the term for a processor's request for a word that is not currently in the cache?</b>                                                                  |
|     | <b>Answer:</b> A cache miss.                                                                                                                                           |
| #53 | <b>Q53: Following a cache miss, what action is taken with the requested data word?</b>                                                                                 |
|     | <b>Answer:</b> The word is brought from the lower-level memory into the cache.                                                                                         |
| #54 | <b>Q54: What is the simplest way to assign a location in the cache for each word in memory?</b>                                                                        |
|     | <b>Answer:</b> To assign the cache location based on the address of the word in memory.                                                                                |
| #55 | <b>Q55: Term: Direct-mapped cache</b>                                                                                                                                  |
|     | <b>Answer:</b> A cache structure in which each memory location is mapped to exactly one location in the cache.                                                         |
| #56 | <b>Q56: What is the typical formula used by direct-mapped caches to determine the block location for a given memory address?</b>                                       |
|     | <b>Answer:</b> (Block address) modulo (Number of blocks in the cache).                                                                                                 |

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| #57 | <b>Q57: In a direct-mapped cache where the number of entries is a power of 2, how is the mapping index computed from the address?</b>                     |
|     | <b>Answer:</b> By using the low-order $\log_2(\text{cache size in blocks})$ bits of the block address.                                                    |
| #58 | <b>Q58: An 8-block direct-mapped cache uses the _____ lowest bits of the block address as the cache index.</b>                                            |
|     | <b>Answer:</b> 3                                                                                                                                          |
| #59 | <b>Q59: How does a cache distinguish between the multiple different memory locations that could map to the same cache entry?</b>                          |
|     | <b>Answer:</b> By using tags, which store the upper portion of the address.                                                                               |
| #60 | <b>Q60: Term: Tag</b>                                                                                                                                     |
|     | <b>Answer:</b> A field in a memory hierarchy table that contains the address information required to identify if a block corresponds to a requested word. |
| #61 | <b>Q61: What portion of the memory address must the tag field in a cache contain?</b>                                                                     |
|     | <b>Answer:</b> The upper portion of the address, corresponding to the bits not used as an index into the cache.                                           |
| #62 | <b>Q62: Why are the index bits of a memory address typically omitted from the cache tag?</b>                                                              |
|     | <b>Answer:</b> They are redundant, as the index field by definition must match the cache block number being accessed.                                     |
| #63 | <b>Q63: What mechanism indicates that a cache block does not contain valid information, such as on processor start-up?</b>                                |
|     | <b>Answer:</b> A valid bit.                                                                                                                               |
| #64 | <b>Q64: Term: Valid bit</b>                                                                                                                               |
|     | <b>Answer:</b> A field in the tables of a memory hierarchy that indicates that the associated block in the hierarchy contains valid data.                 |
| #65 | <b>Q65: If a cache block's valid bit is not set, what is the outcome of a tag comparison for that block?</b>                                              |
|     | <b>Answer:</b> A match is impossible, resulting in a miss.                                                                                                |

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| #66 | <b>Q66: Caching is considered the most important example of what "big idea" in computer architecture?</b>                                                                            |
|     | <b>Answer:</b> Prediction.                                                                                                                                                           |
| #67 | <b>Q67: In the example sequence of memory references, what event occurs on the eighth reference to address 18 (\$10010_{two}\$)?</b>                                                 |
|     | <b>Answer:</b> It causes a miss and must replace the word at address 26 (\$11010_{two}\$) in cache block 2 (\$010_{two}\$).                                                          |
| #68 | <b>Q68: In a direct-mapped cache, what happens if a newly requested item maps to a block that is already occupied by valid data?</b>                                                 |
|     | <b>Answer:</b> The existing data is replaced, as there is only one possible location for the new item.                                                                               |
| #69 | <b>Q69: For a simple cache lookup, a referenced memory address is primarily divided into which two fields?</b>                                                                       |
|     | <b>Answer:</b> A tag field and a cache index.                                                                                                                                        |
| #70 | <b>Q70: What is the function of the cache index derived from a memory address?</b>                                                                                                   |
|     | <b>Answer:</b> It is used to select a specific block (or entry) within the cache.                                                                                                    |
| #71 | <b>Q71: During a cache access, what is the tag field from the memory address used for?</b>                                                                                           |
|     | <b>Answer:</b> It is compared with the tag value stored in the selected cache block to verify an address match.                                                                      |
| #72 | <b>Q72: What two conditions must be met for a cache access to result in a 'hit'?</b>                                                                                                 |
|     | <b>Answer:</b> The tag from the address must match the tag stored in the cache block, and the block's valid bit must be on.                                                          |
| #73 | <b>Q73: The total number of entries in a direct-mapped cache must be a ____.</b>                                                                                                     |
|     | <b>Answer:</b> power of 2                                                                                                                                                            |
| #74 | <b>Q74: For a 64-bit address system with a direct-mapped cache of <math>2^n</math> blocks and a block size of <math>2^m</math> words, what is the size of the tag field in bits?</b> |
|     | <b>Answer:</b> The tag size is $64 - n - (m+2)$ bits.                                                                                                                                |



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| #75 | <b>Q75: What is the formula for the total number of bits in a direct-mapped cache with <math>2^n</math> blocks?</b>              |
|     | <b>Answer:</b> $2^n \times (\text{block size in bits} + \text{tag size in bits} + 1 \text{ for valid bit})$ .                    |
| #76 | <b>Q76: When stating a cache's size (e.g., '16 KiB cache'), what parts are conventionally excluded from the total bit count?</b> |
|     | <b>Answer:</b> The storage for tags and valid bits are excluded; only the data storage is counted.                               |
| #77 | <b>Q77: For a cache with 16 KiB of data and four-word blocks, how many blocks does the cache contain?</b>                        |
|     | <b>Answer:</b> 1024 blocks (16 KiB = 4096 words; $4096 \text{ words} / 4 \text{ words/block} = 1024 \text{ blocks}$ ).           |
| #78 | <b>Q78: To find the block address from a given byte address, what calculation is performed?</b>                                  |
|     | <b>Answer:</b> Block address = $\lfloor \frac{\text{Byte address}}{\text{Bytes per block}} \rfloor$ .                            |
| #79 | <b>Q79: Given a cache with 64 blocks and a 16-byte block size, to what block number does byte address 1200 map?</b>              |
|     | <b>Answer:</b> Block number 11 (since $(1200 / 16) \bmod 64 = 75 \bmod 64 = 11$ ).                                               |
| #80 | <b>Q80: What principle of locality do larger cache blocks primarily exploit to lower miss rates?</b>                             |
|     | <b>Answer:</b> Spatial locality.                                                                                                 |
| #81 | <b>Q81: What is the typical effect on miss rate when increasing the cache block size?</b>                                        |
|     | <b>Answer:</b> It usually decreases the miss rate, up to a certain point.                                                        |
| #82 | <b>Q82: What negative consequence can occur if the block size becomes a significant fraction of the total cache size?</b>        |
|     | <b>Answer:</b> The miss rate may increase due to heavy competition for a smaller number of available blocks.                     |
| #83 | <b>Q83: What is the most significant drawback of increasing cache block size?</b>                                                |
|     | <b>Answer:</b> It increases the miss penalty, as fetching a larger block from memory takes more time.                            |
| #84 | <b>Q84: Term: Cache miss</b>                                                                                                     |
|     | <b>Answer:</b> A request for data from the cache that cannot be filled because the data are not present in the cache.            |

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| #85 | <b>Q85: A cache miss creates a _____, which freezes the processor while waiting for memory, as opposed to a full context-switching exception.</b> |
|     | <b>Answer:</b> pipeline stall                                                                                                                     |
| #86 | <b>Q86: What is the first step in the process of handling an instruction cache miss?</b>                                                          |
|     | <b>Answer:</b> Send the original Program Counter (PC) value to the memory system.                                                                 |
| #87 | <b>Q87: What is the second step in handling an instruction cache miss?</b>                                                                        |
|     | <b>Answer:</b> Instruct main memory to perform a read and wait for the access to complete.                                                        |
| #88 | <b>Q88: What is the third step in handling an instruction cache miss?</b>                                                                         |
|     | <b>Answer:</b> Write the new cache entry, loading data from memory, storing the upper address bits as the tag, and setting the valid bit.         |
| #89 | <b>Q89: What is the final step after a cache miss has been fully handled?</b>                                                                     |
|     | <b>Answer:</b> Restart the instruction execution at the first step, which will refetch the instruction (now from the cache).                      |
| #90 | <b>Q90: If a store instruction updates only the cache and not main memory, the cache and memory are said to be _____.</b>                         |
|     | <b>Answer:</b> inconsistent                                                                                                                       |
| #91 | <b>Q91: Term: Write-through</b>                                                                                                                   |
|     | <b>Answer:</b> A scheme in which writes always update both the cache and the next lower level of the memory hierarchy.                            |
| #92 | <b>Q92: What is the major performance disadvantage of a simple write-through cache scheme?</b>                                                    |
|     | <b>Answer:</b> Every write operation incurs the long latency of writing to main memory, which can significantly slow down the processor.          |
| #93 | <b>Q93: What hardware component is often used to mitigate the performance penalty of a write-through cache?</b>                                   |
|     | <b>Answer:</b> A write buffer.                                                                                                                    |

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| #94  | <b>Q94: Term: Write buffer</b>                                                                                                                                     |
|      | <b>Answer:</b> A queue that holds data while the data are waiting to be written to memory, allowing the processor to continue execution.                           |
| #95  | <b>Q95: Under what condition must a processor stall even when using a write buffer?</b>                                                                            |
|      | <b>Answer:</b> The processor must stall if it needs to perform a write when the write buffer is already full.                                                      |
| #96  | <b>Q96: Term: Write-back</b>                                                                                                                                       |
|      | <b>Answer:</b> A scheme that handles writes by updating values only to the block in the cache, then writing the modified block to memory only when it is replaced. |
| #97  | <b>Q97: In a write-through cache, the strategy of fetching the block from memory on a write miss before overwriting the relevant portion is called ____.</b>       |
|      | <b>Answer:</b> write allocate                                                                                                                                      |
| #98  | <b>Q98: What is the 'no write allocate' policy for a write miss?</b>                                                                                               |
|      | <b>Answer:</b> An alternative strategy where the portion of the block in memory is updated, but the block is not brought into the cache.                           |
| #99  | <b>Q99: Why can't a write-back cache simply overwrite a block on a store instruction before confirming a hit?</b>                                                  |
|      | <b>Answer:</b> Doing so could destroy the only valid copy of the data if the block was modified and has not yet been written back to memory.                       |
| #100 | <b>Q100: What kind of cache organization does the Intrinsity FastMATH processor use to avoid stalling its pipeline?</b>                                            |
|      | <b>Answer:</b> Separate (split) instruction and data caches.                                                                                                       |
| #101 | <b>Q101: The Intrinsity FastMATH processor's data cache is 16 KiB, contains 256 blocks, and has a block size of ____ words.</b>                                    |
|      | <b>Answer:</b> 16                                                                                                                                                  |
| #102 | <b>Q102: For the 32-bit Intrinsity FastMATH cache, what are the respective sizes of the tag, index, and block offset fields?</b>                                   |
|      | <b>Answer:</b> The tag is 18 bits, the index is 8 bits, and the block offset (word + byte) is 6 bits.                                                              |

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| #10<br>3 | <b>Q103: What two write policies does the Intrinsity FastMATH processor support?</b>                                                                                           |
|          | <b>Answer:</b> It supports both write-through and write-back, selectable by the operating system.                                                                              |
| #10<br>4 | <b>Q104: Term: Split cache</b>                                                                                                                                                 |
|          | <b>Answer:</b> A scheme where a memory hierarchy level has two independent caches operating in parallel, one for instructions and one for data.                                |
| #10<br>5 | <b>Q105: What is the primary motivation for using split instruction and data caches in modern processors?</b>                                                                  |
|          | <b>Answer:</b> To increase cache bandwidth to meet the high demands of modern pipelines, allowing simultaneous instruction and data access.                                    |
| #10<br>6 | <b>Q106: How does the miss rate of a 32 KiB combined cache typically compare to the effective miss rate of two 16 KiB split caches?</b>                                        |
|          | <b>Answer:</b> The combined cache usually has a slightly lower (better) miss rate.                                                                                             |
| #10<br>7 | <b>Q107: Why is the slightly higher miss rate of a split cache an acceptable trade-off?</b>                                                                                    |
|          | <b>Answer:</b> The advantage of doubling the cache bandwidth far outweighs the small disadvantage of a slightly increased miss rate.                                           |
| #10<br>8 | <b>Q108: Using a cache block size larger than one word improves efficiency by reducing the amount of _____ storage relative to data storage.</b>                               |
|          | <b>Answer:</b> tag                                                                                                                                                             |
| #10<br>9 | <b>Q109: Which design guideline is generally valid regarding cache block size and memory bandwidth?</b>                                                                        |
|          | <b>Answer:</b> The higher the memory bandwidth, the larger the cache block can be.                                                                                             |
| #11<br>0 | <b>Q110: What technique, called _____, reduces the effective miss penalty by resuming execution as soon as the specifically requested word of a block arrives from memory?</b> |
|          | <b>Answer:</b> early restart                                                                                                                                                   |
| #11<br>1 | <b>Q111: What memory transfer technique organizes the memory to send the requested word first, then the rest of the block, to reduce miss penalty?</b>                         |
|          | <b>Answer:</b> Requested word first (or critical word first).                                                                                                                  |